

MASSEY'S RANKING

AP

RATING AND RANKING

MOTIVATIONS

the ability to

- *rate* something (is this a warm day for June in London?), or to
- *rank* a set of elements (which is the coldest day of the month?)

is part of Science and Engineering since before Data Science.

Rating & ranking is a good framework to introduce Data Science techniques of general value and wide applicability.

Sports R&R is both fun and a huge Data Science market!

DEFINITION

A measure of value of the subject, as objective and replicable as possible.

E.g., temperature.

Normally, abilities are

- latent
- hard to measure
- time-dependent
- place-dependent

Exercise: take the [Prof or Hobo?](#) quiz!

yet, abilities are also

- hard to *transcend* (revert-to-the-mean effect, RTTM)
- relatively easy to perceive and project

EXAMPLE: FOOTBALL

- hard to guess the single score $\implies$ entertainment value
- easy for experts to guess the long-term effect $\implies$ different levels of enjoyment; RTTM: Revert To The Mean effect

Low scoring creates **randomness**

FOR REFERENCE: FORMALISATION

1-DIMENSIONAL RANKING

P : players, $|P| = n$

T : time instants

$r : P \times T \rightarrow \mathbb{R}$

A given rating function r creates a ranking (ρ) on a set:

$$\rho : P \times T \rightarrow [1..n]$$

$$\rho(p_i, t) = k \leftrightarrow |\{p_j : r(p_j, t) \leq r(p_i, t)\}| = k$$

$$\delta(p_i, p_j, t) = |r(p_i, t) - r(p_j, t)|$$

δ captures both similarity and distance

MULTI-DIMENSIONAL RANKING

Multi-dim. rating:

$$r_{multi} : P \times T \rightarrow \mathbb{R}^d$$

Often:

$$r_{multi}(p_i, t) : f(r_1(p_i, t), \dots, r_d(p_i, t))$$

Pareto dominance:

p_i dominates p_j (at time t) if on every dimension x

$$r_x(p_i, t) \geq r_x(p_j, t)$$

RATING IN GAMES

RATINGS IN GAMES

- score-based games are better-suited to create ratings
- yet effect of time and hardness of the proposed test match could be hard to assess.

SHOULD GAMES KEEP USER RATINGS?

YES:

- feeling of improvement
- a gauge for new features
- leads to rankings:
 - better matchmaking $\implies$ entertainment value
 - fraud/anomaly detection?

NO: GAME PROWESS AS SOCIAL RANKING?

The spectacle is a social relation mediated by images, not a collection of images.

«Le spectacle n'est pas un ensemble d'images, mais un rapport social entre des personnes, médiatisé par des images»

[Guy Debord, La Société du spectacle (1967), Thèse 4]

- a reflection of US culture?
- a turn-off for people who don't feel competitive?
- turns-off casual users?

SPORT RANKING/ESTIMATION

DOMAIN

- n teams play each other in a tournament
- final scores are recorded, e.g., Real Madrid–Borussia Dortmund: 2-0.

Focus on predicting

- the winner of a future match
- the score difference (eg, $2-0=2$)

RUNNING EXAMPLE

NCAA Atlantic Coast Conference

- win-loss balance and the points balance are second-level performance measures.
- they are not considered sufficient to create valuable ratings/rankings/predictions.

	Duke	Miami	UNC	UVA	VT	Record	Point Differential
Duke		7-52	21-24	7-38	0-45	0-4	-124
Miami	52-7		34-16	25-17	27-7	4-0	91
UNC	24-21	16-34		7-5	3-30	2-2	-40
UVA	38-7	17-25	5-7		14-52	1-3	-17
VT	45-0	7-27	30-3	52-14		3-1	90

Ranking *blurs* rating: 3rd has has a much lower rating than the 2nd.

[IN]CREDIBLE ASSUMPTIONS

1. to each team a **latent** variable for *strength* is assigned

numerical **ratings** determine a **ranking** among teams (at $t=\text{end}$, so we can drop it)

and a prediction $Pr[a \rightarrow b] = \frac{\rho(a)}{\rho(a) + \rho(b)}$

2. strength/rating is immutable during the tournament
3. teams play each other exactly once during the tournament

Now, consider the score difference in each match, say i vs. j , defined as

$$s_i - s_j$$

Define $\mathbf{y}_{m \times 1}$ as the vector of all score differences in matches

Assume (assumption 4) that strength/rating imbalance determines score difference:

$$r_i - r_j = s_i - s_j$$

$$X_{m \times n} \cdot \mathbf{r}_{n \times 1} = \mathbf{y}_{m \times 1}$$

$$\begin{bmatrix} 0 & 0 & +1 & 0 & -1 & 0 \\ 0 & \ddots & \ddots & & & \\ \ddots & \ddots & \ddots & \ddots & & \\ \ddots & \ddots & \ddots & \ddots & \ddots & \\ & \ddots & \ddots & \ddots & \ddots & \\ 0 & -1 & 0 & +1 & 0 & 0 \end{bmatrix}$$

$X_{m \times n}$ with $m \gg n$ is over-constrained, no hope of finding a solution.

MASSEY'S RATINGS

DATA PREPARATION

Massey considered the equivalent formulation of

$$X_{m \times n} \cdot \mathbf{r}_{n \times 1} = \mathbf{y}_{m \times 1}$$

as

$$X^T \cdot X \cdot \mathbf{r} = X^T \cdot \mathbf{y}$$

Both sides are easier to work with.

Right-hand side: $X^T \cdot \mathbf{y}$ is the all-season points difference vector, called $\mathbf{p}$.

Notice that $\sum p_i = 0$.

On the left-hand side,

$$M_{n \times n} = X^T X$$

is squared, semidefinite and positive.

However, the rows sum to 0 and cols. are not independent: 0/ ∞ solutions ensue...

M. also noticed that M has a fixed structure and does not need to be re-computed all the times.

$$\begin{bmatrix} n-1 & 0 & -x & 0 & -y & 0 \\ & n-1 & \ddots & \ddots & & \\ & & \ddots & \ddots & \ddots & \\ & & & \ddots & \ddots & \\ & & & & \ddots & \\ 0 & -z & 0 & -w & 0 & n-1 \end{bmatrix}$$

$m_{i,i} = n - 1$ is the numbers of games i played,

$m_{i,j}$ is the negation of the no. of matches between i and j : all values are set to -1.

$$\begin{pmatrix} 4 & -1 & -1 & -1 & -1 \\ -1 & 4 & -1 & -1 & -1 \\ -1 & -1 & 4 & -1 & -1 \\ -1 & -1 & -1 & 4 & -1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{pmatrix} = \begin{pmatrix} -124 \\ 91 \\ -40 \\ -17 \\ 0 \end{pmatrix}$$

MASSEY

1. drops the last row/match
2. replaces it with a row of 1s, and sets $p_n = 0$

(all ratings, positive and negative, will sum to 0)

$\overline{M} = M$ everywhere but for the last row which is full of 1s

$\overline{\mathbf{p}}$ is $\mathbf{p}$ everywhere but for the last el. $p_n = 0$.

1. now $\overline{M}$ is non-singular and invertible

2. solves

$$\overline{M}\mathbf{r} = \overline{\mathbf{p}}$$

to obtain an approximated rating for the teams.

The MSE solution to Massey's formula is a form of **regression**.

It can also be seen as $\mathbf{r} = (\overline{X^T X})^{-1} \overline{X^T \mathbf{y}}$.

OUTPUT

ratings sum to zero

values have no direct interpretation.

however, they effectively generate a hierarchy.

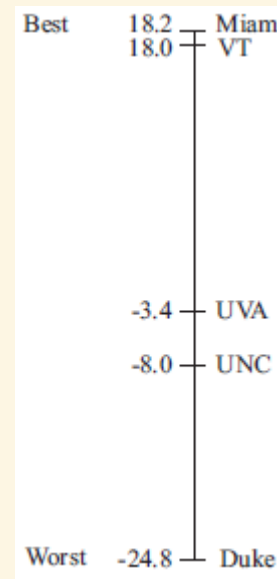
Team	Rating r	Rank
Duke	-24.8	5th
Miami	18.2	1st
UNC	-8.0	4th
UVA	-3.4	3rd
VT	18.0	2nd

VISUALISE THE RANKING

Ratings are not necessarily meaningful, as a result of the matrix preparation.

Rankings are meaningful:

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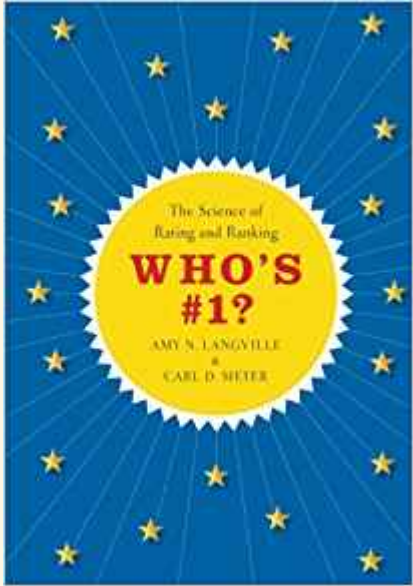


CONCLUSIONS

POINTS TO FOCUS ON

- rating and rating is the fun side of Data Science!
- *latent* variables that represent *non-measurable* skills
- those leave in a ***feature space*** possibly separated from the ***data space***
- yet they may get a numeric estimate, and inform our predictions
- Massey regresses on the latent variables

FURTHER READINGS



[...] overview of the mathematical algorithms and methods used to rate and rank sports teams, political candidates, products, Web pages, and more.

They survey and compare the different methods employed today, showing why their strengths and weaknesses depend on the underlying goal, and explaining why and when a given method should be considered.

[...] describe what can and can't be expected from the most widely used systems.